

Hardy Wang

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TECHNICAL SKILLS

Computer: C/C++, JavaScript/Typescript, MySQL, Python, MATLAB, Arduino, System Verilog, RISC-V Assembly
Software: Git, Google Colab, VS Code, SolidWorks, STM32CubeIDE, Altium designer, AutoCAD, LTSpice, FreeRTOS
Electrical: PCB Design/Debugging, Wire Harnessing, Power Electronics, THT/SMD Soldering
Tools: Multimeters, Oscilloscopes, Function Generators, Bench Power Supply, Circuit Design/Components
Communication: CAN, SPI, I2C, UART

EDUCATION

University of British Columbia

Bachelor of Applied Science in Electrical Engineering

Vancouver, BC

Sept. 2024 – Present

EXPERIENCE

Hardware Engineering Intern

May 2026 – Present

e-Radio inc.

Toronto, ON

- Designed a fleet-deployable embedded sensor node (ESP32, RF receiver, RTC, SD storage)
- Designed the RF front end, selecting the antenna and its impedance-matching network, and isolated the receiver from high-noise nets on the board — the I2C bus, the WiFi module, and other digital switching — rather than relying on shielding alone
- Diagnosed and resolved an ADC/WiFi radio resource conflict at the SoC level, routing analog sensing to ADC and firmware-gating RF transmission to eliminate noise coupling into sensitive analog signal paths.
- Completed schematic-to-layout PCB design and routing for two-layer board, released for fab and SMT assembly
- Contributed firmware support for board bring-up and diagnostics alongside the hardware design

Power and Signals Designer

Sept 2025 – Present

UBC Solar Car

Vancouver, BC

- Designing and testing PCB assemblies for the vehicle's electrical and control systems, with a focus currently on the Steering wheel board, responsible for majority of driver input including CAN enabled functions
- Utilizing Altium Designer and hardware debugging tools to iterate board layouts, validate circuit performance, and ensure compliance with system-level power and safety requirements
- Created integration meetings with relevant teams to ensure all requirements were satisfied and board development deadlines were met
- Assembled and hand-soldered PCB prototypes and final boards used in the vehicle, ensuring reliable electrical connections and readiness for integration
- Created and managed the PCB Bill of Materials (BOM), optimizing component selection for cost

Project Technician Intern

May 2026 – Present

Aecon Group Inc.

Toronto, ON

- Collected, cleaned, and validated engineering design data to support the training of a machine learning model for automated Bill of Materials (BOM) generation.
- Created and reviewed electrical Bills of Materials (BOMs), verifying component selections and ensuring consistency with engineering design standards and project specifications.

PROJECTS

Steering Wheel PCB | Altium Designer

Dec 2025 – May 2026

- Designed a steering wheel interface PCB in Altium Designer, implemented differential pair routing for CAN bus communication and discrete driver inputs including horn, push-to-talk, regenerative braking, and display controls.
- Led schematic design, component selection, and PCB layout, optimizing board size, connector orientation, and switch placement to match updated steering wheel geometry
- Updated MCU programming and debug support from ST-Link to JTAG, ensuring proper STM32 pin routing, signal integrity, and test access.
- Mitigated noise and EMI through opto-isolation and dedicated return paths, protecting sensitive MCU logic from high-transient signals